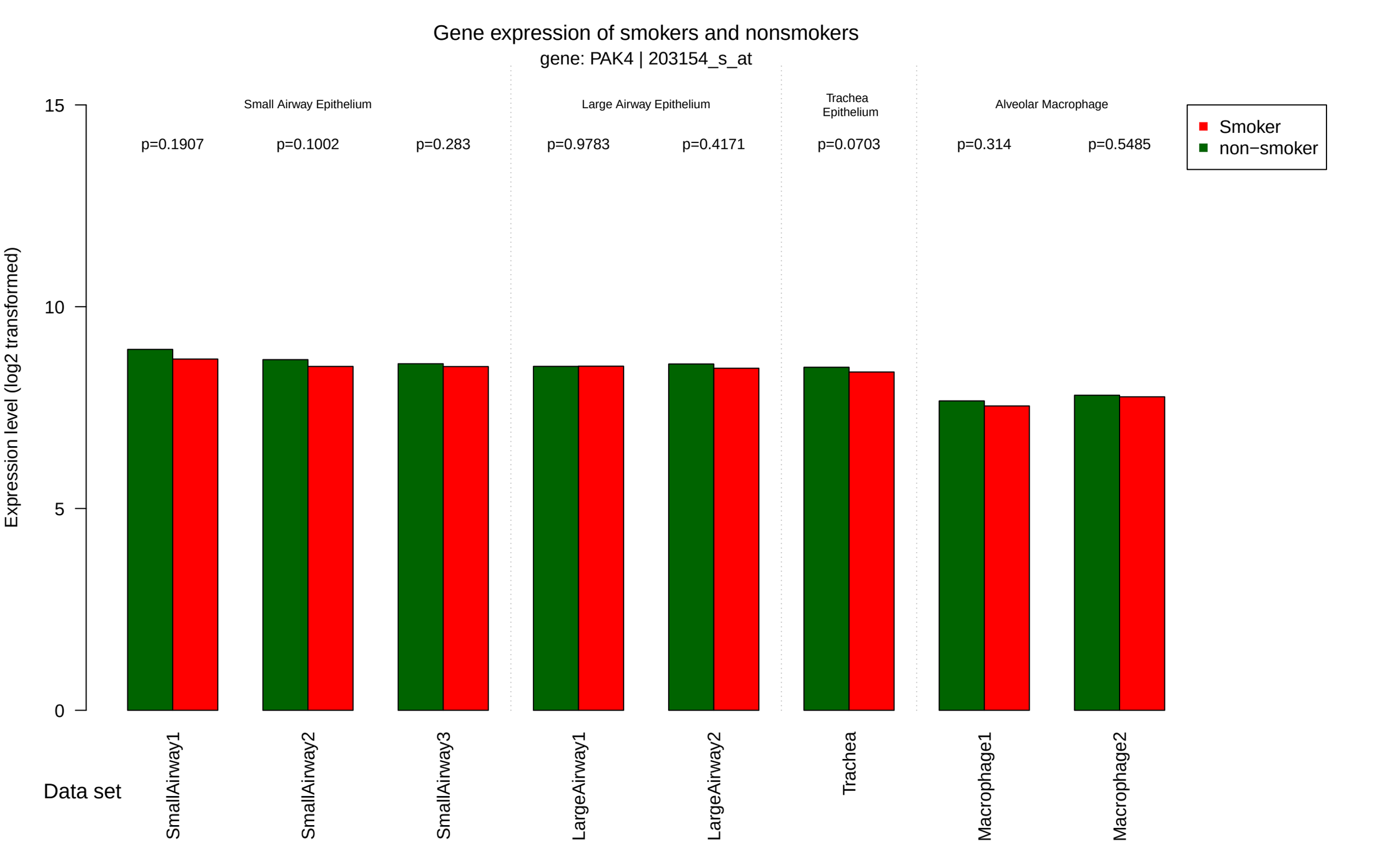
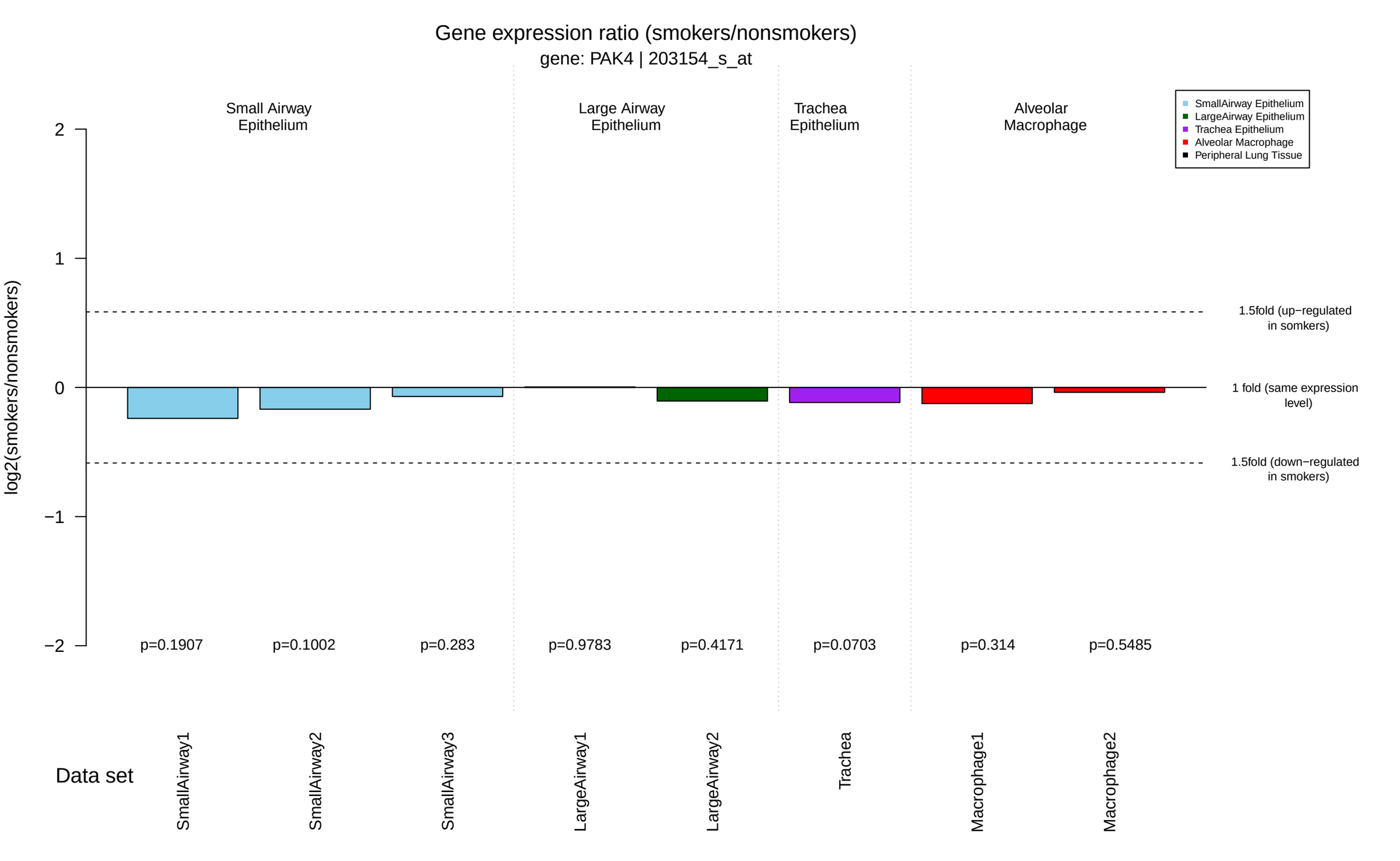
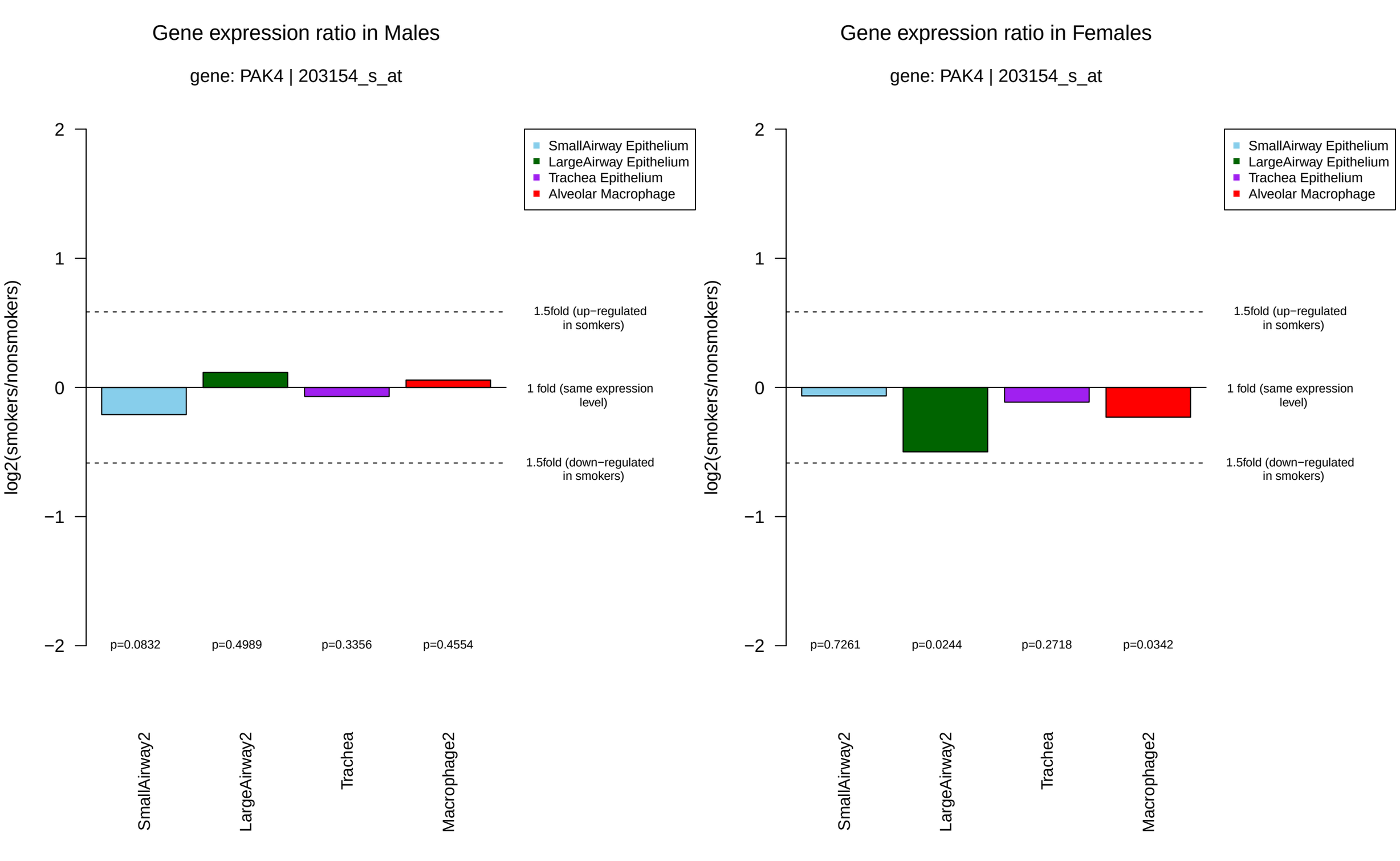
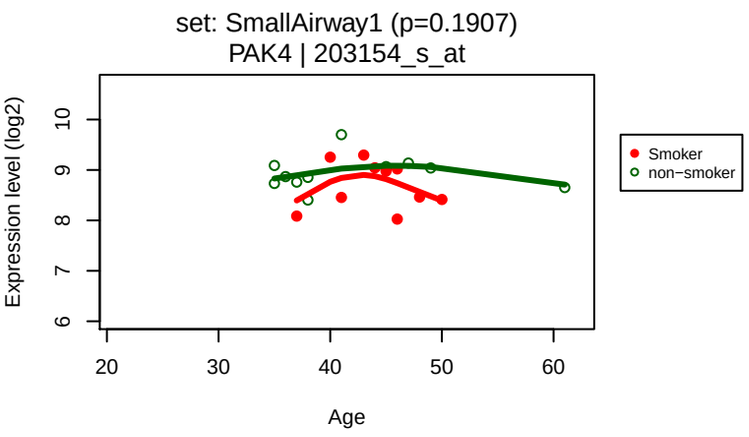
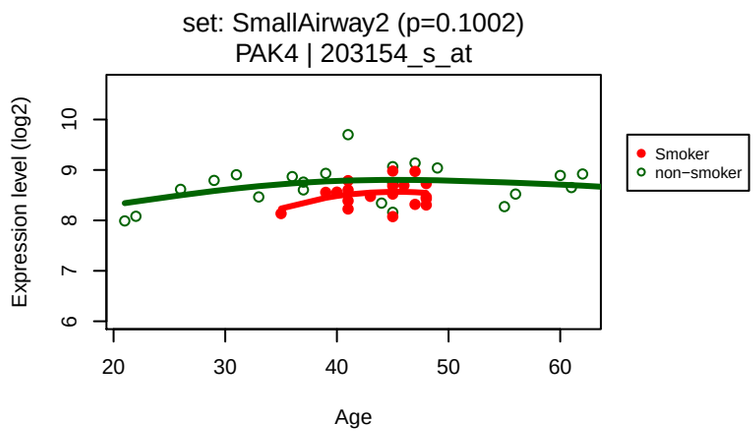
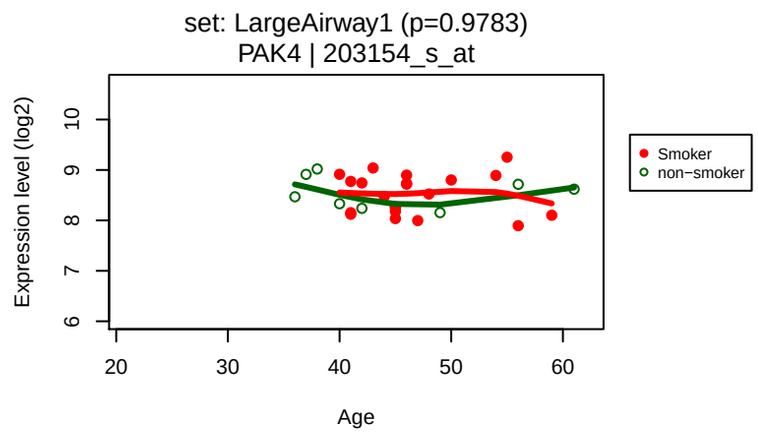

Gene expression of smokers and nonsmokers
gene: PAK4 | 203154_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1907
p=0.1002
p=0.283
p=0.9783
p=0.4171
p=0.0703
p=0.314
p=0.5485
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

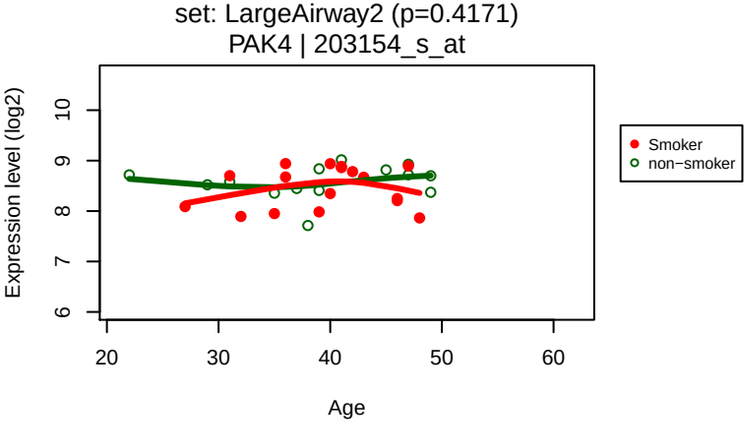

Gene expression ratio (smokers/nonsmokers)
gene: PAK4 | 203154_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1907
p=0.1002
p=0.283
p=0.9783
p=0.4171
p=0.0703
p=0.314
p=0.5485
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

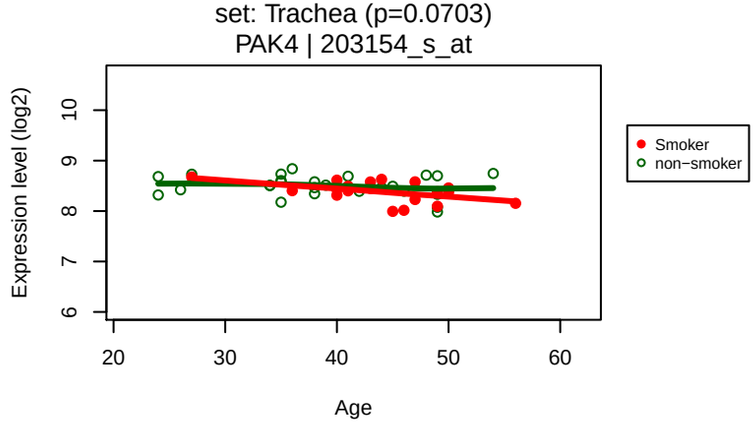

Gene expression ratio in Males
gene: PAK4 | 203154_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0832
p=0.4989
p=0.3356
p=0.4554
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK4 | 203154_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7261
p=0.0244
p=0.2718
p=0.0342
SmallAirway2
LargeAirway2
Trachea
Macrophage2

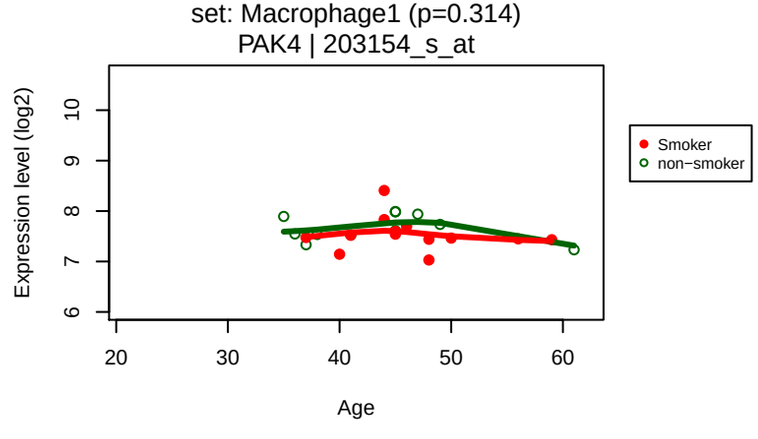

set: SmallAirway1 (p=0.1907)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker

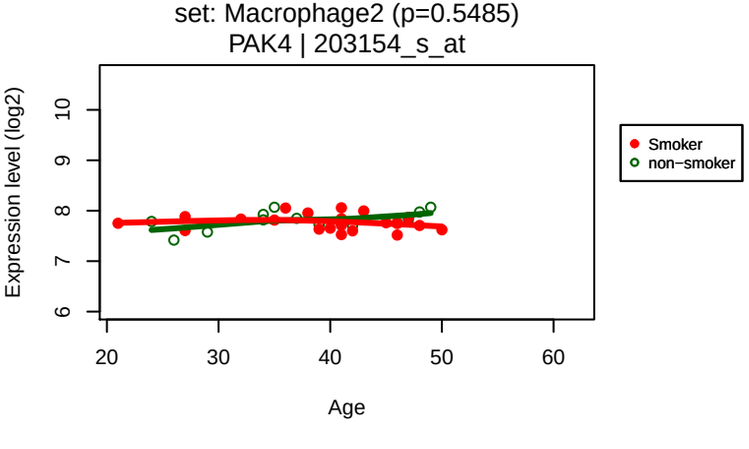

set: SmallAirway2 (p=0.1002)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker

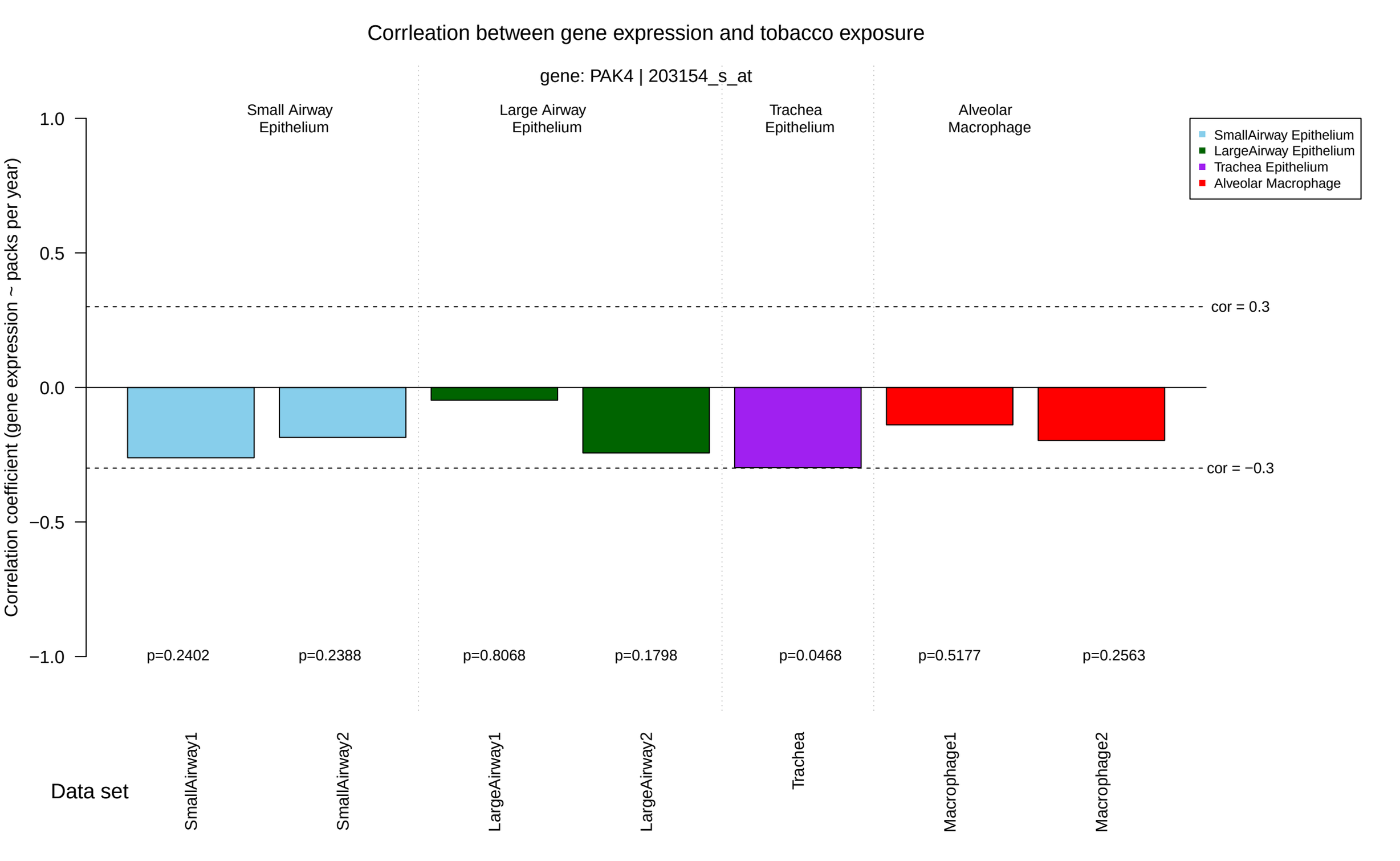

set: LargeAirway1 (p=0.9783)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4171)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0703)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.314)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5485)
PAK4 | 203154_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK4 | 203154_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2402
p=0.2388
p=0.8068
p=0.1798
p=0.0468
p=0.5177
p=0.2563
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2